Unit III Assignment

1. **Counter using useState**

```jsx
import React, { useState } from "react";

function Counter() {
    const [count, setCount] = useState(0);

    return (
        <div>
            <h2>Count: {count}</h2>

            <button onClick={() => setCount(count + 1)}>
                Increment
            </button>

            <button onClick={() => setCount(count - 1)}>
                Decrement
            </button>
        </div>
    );
```

2. **Fix Infinite Re-render Loop**

   setCount() ko render ke andar directly call karne se infinite re-render hota hai. Isliye button click par stateupdate karna chahiye.

```jsx
import React, { useState } from "react";

function Counter() {
    const [count, setCount] = useState(0);

    return (
        <div>
            <p>Count: {count}</p>

            <button onClick={() => setCount(count + 1)}>
                Increment
            </button>
        </div>
    );
}

export default Counter;
```

3. **useEffect to Fetch Data**

```jsx
        .then(response => response.json())
        .then(data => {
            setPosts(data);
        })
        .catch(error => {
            console.error("Error:", error);
        });
}, []);

return (
    <div>
        <h2>Posts</h2>

        {posts.map(post => (
            <p key={post.id}>{post.title}</p>
        ))}
    </div>
);
```

4. Lift Shared State Up

Dono sibling components ke paas alag-alag price state thi. State ko parent component mein rakhkar props ke through children ko pass karenge.

```
    return <p>Price: {price}</p>;
}

function App() {
    const [price, setPrice] = useState(0);

    return (
        <div>
            <h2>Product Price</h2>

            <PriceInput
                price={price}
                setPrice={setPrice}
            />

            <PriceDisplay price={price} />
        </div>
    );
```

4. **Custom useLocalStorage Hook**

```
import { useState } from "react";

function useLocalStorage(key, initialValue) {

    const [value, setValue] = useState(() => {
        const savedValue = localStorage.getItem(key);

        return savedValue
            ? JSON.parse(savedValue)
            : initialValue;
    });

    const setStoredValue = newValue => {
        setValue(newValue);
        localStorage.setItem(key, JSON.stringify(newValue));
    };

    return [value, setStoredValue];
```

```jsx
function App() {
    const [theme, setTheme] = useLocalStorage("theme", "light");

    return (
        <div>
            <h2>Theme: {theme}</h2>

            <button onClick={() => setTheme("light")}>
                Light
            </button>

            <button onClick={() => setTheme("dark")}>
                Dark
            </button>
        </div>
    );
```

6. Fix React Key Warning

React list items ko unique key deni chahiye.

```
{items.map(item => (
  <li key={item.id}>
    {item.name}
  </li>
))}
```

7. **Reusable Button Component**

```jsx
import React from "react";
import "./Button.css";

function Button({ variant = "primary", children }) {
    return (
        <button className={`btn ${variant}`}>
            {children}
        </button>
    );
}

export default Button;
```

CSS

```css
.btn {
    padding: 10px 20px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    color: white;
}

.primary {
    background-color: blue;
}

.danger {
    background-color: red;
}
```

Usage

```jsx
<Button variant="primary">
    Save
</Button>

<Button variant="danger">
    Delete
</Button>
```

8. Fix ChildCard Re-render Using React.memo

Parent component ke re-render hone par normally uske child components bhi re-render hote hain, even when their props have not changed. React.memo unnecessary re-render ko prevent kar sakta hai.

```
    return <h3>{title}</h3>;
});

function ParentDashboard() {
    const [tick, setTick] = useState(0);

    return (
        <div>
            <button onClick={() => setTick(tick + 1)}>
                Refresh
            </button>

            <p>Refresh Count: {tick}</p>

            <ChildCard title="Static Title" />
        </div>
    );
```

9. React Router v6 – 3 Routes

Required routes: /, /products, and /products/:id

10. Single State Object with Shared onChange

Original code mein har field ke liye separate useState tha. Isko ek state object mein combine kar sakte hain.

```
import React, { useState } from "react";

function ContactForm() {

    const [formData, setFormData] = useState({
        name: "",
        email: "",
        phone: ""
    });

    const handleChange = (e) => {
        const { name, value } = e.target;

        setFormData({
            ...formData,
            [name]: value
        });
    };
```

```
          [name]: value
      });
  };

  const handleSubmit = (e) => {
      e.preventDefault();

      console.log(formData);
  };

  return (
      <form onSubmit={handleSubmit}>

          <input
              type="text"
              name="name"
              placeholder="Enter Name"
              value={formData.name}
```

```jsx
<input
    type="email"
    name="email"
    placeholder="Enter Email"
    value={formData.email}
    onChange={handleChange}
/>

<input
    type="tel"
    name="phone"
    placeholder="Enter Phone"
    value={formData.phone}
    onChange={handleChange}
/>

<button type="submit">
    Submit
```